Secure Consensus Mechanisms for Blockchained IoT: A Review

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***Abstract*—The inherent security features of distributed ledger technologies can offer a promising solution to enhance the security of IoT systems, addressing their inevitable security concerns. Nonetheless, it is essential to consider potential flaws, particularly those related to consensus mechanisms within these blockchain-based IoT systems, to ensure the overall security and integrity of the network. Any threat at the blockchain level could potentially harm the entire IoT network. This paper investigates the security of the underlying consensus mechanisms within blockchain-based IoT systems. We present a review of securing consensus mechanisms within the Internet of Blockchained Things.**

***Index Terms*—IoT, Blockchain, Consensus, Attacks, Security**

I. Introduction

The internet of things (IoT) is a network of interconnected devices equipped with sensors that can sense, collect, transmit and share data with their environment, a central server, or other connected devices over the internet. The applications of IoT, leveraging internet connectivity, range from smart homes and wearable health monitors to industrial automation and smart cities. However, despite its remarkable capabilities in various fields, it still faces many security issues and risks that must be addressed. For example, managing the large volumes of data it generates is challenging. Furthermore, data collected and shared can include sensitive personal information, which requires enhanced security mechanisms to prevent leakage and exploitation by malicious parties. IoT architecture is typically centralized, making it vulnerable to a Single Point of Failure (SPOF) and more prone to attacks [1]. Besides, there are also data privacy concerns which arise due to the lack of distributed control and the concentration of data, potentially leading to issues such as unauthorized data access and breaches. Future improvements in IoT applications face challenges due to these issues. Researchers have opened up the field of integrating decentralized, secure, and transparent Distributed Ledger Technology (DLT), like Blockchain, with IoT to tackle these problems.

Blockchain serves the security of IoT by ensuring privacy through cryptographic algorithms, reliability via tamper-resistant ledgers, and eliminating the SPOF by decentralizing the network [2]. Additionally, the consensus mechanism ensures data integrity and enhances the resilience of IoT networks against various attacks. However, integrating these technologies poses several challenges, including issues related to energy efficiency, computational power, and storage. Furthermore, the complexity of implementation and the incompatibility of IoT devices with different consensus algorithms represent significant obstacles in combining these technologies [3]. Therefore, an adaptation of these mechanisms is necessary to suit IoT limitations. Moreover, these algorithms must be secured against potential threats because malicious actors can undermine them. If attackers gain control over the process of generating blocks, they can jeopardize the integrity of the entire blockchain and its decentralized characteristics.

This paper provides an overview on blockchain technology and its consensus mechanisms, with a focus on securing the consensus mechanisms of blockchained Internet of Things.

The rest of the paper is organized as follows: Section II gives an overview of blockchain technology. Section III explores the concept of the Internet of Blockchained Things. Then, section IV discusses the security issues and threats targeting existing consensus mechanisms. Section V provides a review of secure consensus mechanisms in the Blockchained IoT. A conclusion and future works are presented in Section VI.

II. Blockchain Technology Overview

After the introduction of blockchain technology in Satoshi Nakamoto's Bitcoin whitepaper [4] and the official launch of the blockchain and the Bitcoin network, researchers have increasingly shown interest in its adoption for IoT applications. This section aims to give an overview of this technology.

A. Definition and Key Features

Blockchain technology is a DLT that structures transactions in an immutable chain of blocks. Each of these latter comprises a group of transactions conducted at a particular point in time [5]. This mechanism is decentralized by nature, eliminating the requirement for a third-party trust entity since it operates on a peer-to-peer network [6].

The blockchain is transparent and accessible to anyone, providing a complete historical record of all transactions from the system's inception. Any entity can examine and verify these transactions at their convenience [7]. Blocks in a blockchain are connected through a reference to the previous block,

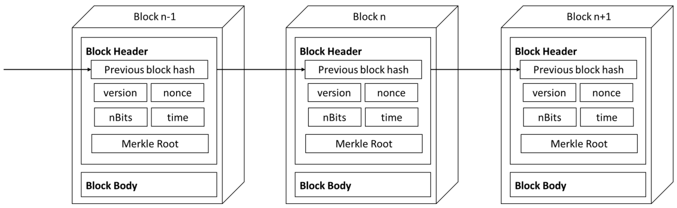

Fig. 1. Blockchain structure: each block contains block header and block body. Blocks are linked by including the hash value of the previous block in each new block's header

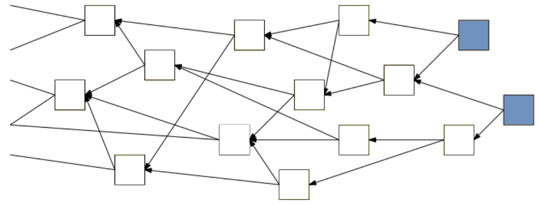

Fig. 2. Snippet from a Tangle: Each transaction (represented by a square) approves two existing transactions (connected by arrows). Blue squares are tips.

each block contains a hash value pointing to its immediate predecessor. The very first block within a blockchain is termed the genesis block and does not have a parent block. Fig 1 shows how blocks are linked together in the blockchain.

This technology guarantees data security and integrity through the use of cryptographic techniques. Besides, it uses the consensus in which the network agrees on the current state of the chain even within a network where participants might not have trust in each other [8]. So, through which all the network agrees on updating the ledger. Furthermore, it offers immutability since once data is recorded in a blockchain, it is almost impossible to alter, tamper with or delete, making it a reliable source. There is also the concept of a smart contract, a computer program that is stored inside of a blockchain, which organizes and manages agreements between network members. When certain predefined conditions are satisfied, the smart contract is activated automatically [7].

B. Consensus Mechanisms

Consensus algorithms enable blockchain to function in a distributed and decentralized manner. They consist of rules and protocols that enable all blockchain nodes to agree on the state of the chain without relying on a trusted third party or central intermediaries, thus, they do not rely on trust. These mechanisms are crucial for maintaining the security of the network. Consensus algorithms vary according to the variety of demands. Some of the popular ones are the following.

Proof of Work: PoW consensus is a process in which nodes (miners) compete to find a hash value of the next block that meets a specified set of criteria, by solving challenging mathematical puzzles. It requires substantial computational power and time. This process deters malicious nodes from easily manipulating the network. However, this consensus is susceptible to 51% attacks, where a single entity or group gains control of over half of the network's mining power, posing a risk to transaction integrity. It is also threatened by a double-spending attack carried out through racing [9].

Proof of Stake: Validators are chosen based on the amount of cryptocurrency they hold and stake, in contrast to PoW, this consensus does not rely on significant computational power [10]. Therefore, the node with the most stake typically becomes the validator. When it succeeds in validating the block, it will be rewarded. On the contrary, if it fails, there is no reward, and some staked cryptocurrency may be lost. For this, executing an attack would be much more expensive. PoS is criticized for being susceptible to nothing-at-stake attacks.

Delegated Proof of Stake: It is a variation which differs from PoS in that the method of selecting validators is through voting. Stakeholders use their coins to elect delegates responsible for consensus. Nodes with the most votes become validators [11]. DPoS is faster and more scalable compared to PoS, but it can be criticized for centralization due to the limited number of validators. Also, it is susceptible to Sybil attacks where an attacker creates multiple fake identities to gain voting power and influence the network.

Practical Byzantine Fault Tolerance: To reach consensus and add the new block in the PBFT algorithm, requires more than two-thirds of all nodes to agree upon that block. Its lower complexity and low energy consumption make this method favourable to private blockchains where third-entity controls the network. However, due to its limited scalability it is considered unsuitable for public blockchains [2].

Proof of Burn: In PoB [12], participants intentionally burn some of their cryptocurrency tokens to demonstrate a commitment to the network, by sending them to a generated address where the money is irrevocably destroyed. In return, participants earn the right to mine or validate transactions. This consensus can be compromised by a double burn attack where an attacker sends the same coins to multiple unspendable addresses, claiming to have burned more than he did.

Proof of Activity: This hybrid consensus algorithm is based on Proof of Work and Proof of Stake [13]. First, miners perform proof of work to create an empty block header with only essential data. Then, validators, based on proof of stake, are chosen to sign the new block and add transactions.

C. IOTA Tangle

IOTA is a DLT specifically designed for IoT. The main idea here is rather than adding blocks, a Directed Acyclic Graph (DAG) structure called *Tangle* is continuously updated with transactions, as shown in Fig 2. Each participant who makes a new transaction has to approve two existing transactions. This concept helps to secure the network and achieve consensus.
The Tangle solves scaling problems as the IoT network grows, by handling transactions in parallel instead of one after another [14]. One crucial aspect of IOTA's protocol is its *Tips Selection Algorithms (TSA)*. *Tips* are transactions not validated yet by incoming ones. Based on TSA, it will be decided which tips

or transactions are chosen to be validated. The most widely used IOTA TSAs are the following.

Uniform Random Tip Selection: URTS is a selection of tips uniformly at random. It gives the same opportunity to each tip to be selected. This makes the selection of the next transaction unpredictable for attackers. But attacks like splitting or parasite chain are easy to perform with this TSA [15].

Markov Chain Monte Carlo: The MCMC algorithm uses a probabilistic approach to select a transaction based on its cumulative weights via random walkers. This latter favours higher cumulative weights so the α parameter was set to control this bias [16]. This algorithm is more computationally expensive than URTS. In case an attacker possesses significantly more computing power, it can be challenging to stop a double-spending attack [17].

III. Internet of Blockchained Things

The main goal of integrating blockchain into the IoT is to meet its growing need for better security, privacy, and decentralization. In a blockchain environment, devices can operate securely and independently without relying on a central authority, which helps reduce risks like SPOF. Additionally, blockchain tracks data during its journey from its origin to its destination and prevents tampering with it. By which, it guarantees the improvements of data integrity, reliability and efficiency for the IoT. Overall, the integration of blockchain and IoT serves the security, reliability and efficiency of IoT.

The applications of blockchain in IoT varies to many domains, as Fig 3 shows, such as smart homes, smart cities, healthcare, agriculture, supply chain management and so on. However, applying it is not straightforward, as it entails various challenges. For instance, the limited processing power and storage capacity of IoT devices. This constraint impacts the feasibility of implementing blockchain protocols. Additionally, traditional blockchain consensus like PoW and PoS are resource-intensive and may not be suitable for IoT. Therefore, adaptations are necessitated to accommodate the requirements of low-power devices. Moreover, latency issues arising from blockchain's consensus can hinder real-time communication in IoT applications that demand instantaneous data processing. Scalability remains another challenge, as blockchain networks must handle an exponentially growing number of IoT devices and transactions without impacting performance.

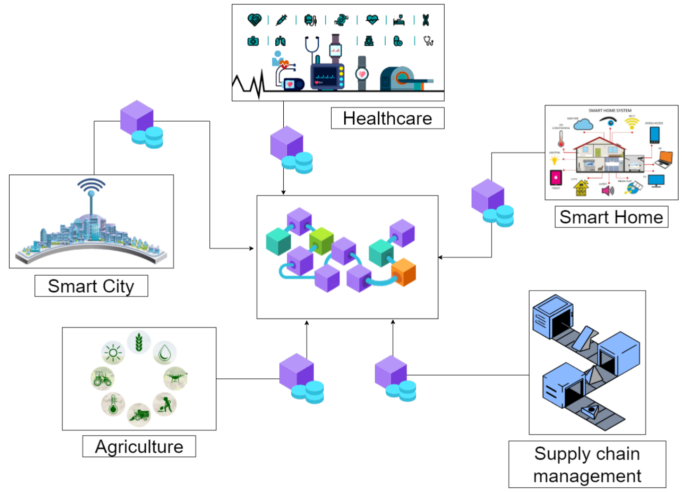

Fig. 3. Internet of Blockchained Things Applications

IV. Security Issues and Threats Targeting Existing Consensus Mechanisms

While consensus algorithms provide a foundational level of security and can serve as effective security mechanisms, they have vulnerabilities. These weaknesses can be exploited by malicious actors to compromise the blockchain's integrity.

As discussed in [22], consensus mechanisms can face both safety and security issues. Safety issues relate to abnormal operations or system crashes during the agreement process among distributed nodes. For example, when honest nodes agree on erroneous data provided by dishonest ones, or when disagreement among honest nodes occurs due to faulty communication. These issues affect two key safety properties: validity and agreement [23] [24], leading to undesirable events such as forks in the chain, double spending, and Sybil attacks.

Security issues are the attacks on the consensus that can be targeting either the consensus nodes (miners, validators, etc) or the main chain consensus [22]. Bharat Bhushan et al. [9] classified these security threats into four categories: double spending threats, network threats, mining pool threats, and wallet security threats. The weaknesses of consensus mechanisms extend beyond those mentioned above. These systems can be vulnerable to various other threats, some of which we enumerate below.

A. 51% Attack

This attack occurs when one or a group of colluding participants possess more than 50% of the network's total power, enabling them to potentially dominate it. In PoW, for example, if attackers control over 51% of the computational power, they can generate blocks faster, build a longer chain, and release it as the main chain. This leads to the original chain being discarded, as miners adopt the longer chain, resulting in transaction reversals even after they have been confirmed. This attack can be exploited for double-spending as well. For instance, attacker A pays seller B for goods. B sends the goods to A. A then buys additional goods from C and uses his computational power to quickly mine a new chain containing the transaction 'A pays C', thereby discarding the first transaction. PoS consensus is less vulnerable to this attack [22] [25]. However, if an attacker gains control of more than 51% of the staked tokens, he can dominate block creation by including and excluding transactions. Delegated PoS is a voting-based consensus. A delegator can benefit from 51% voting power from colluding nodes. Alternatively, delegators can collude to approve conflicting transactions.

B. Sybil Attack

The attacker creates multiple fake identities to influence the network and use them to manipulate the consensus or launch other distributed attacks. They can disrupt communication among nodes by refusing to forward messages or by transmitting incorrect information. Additionally, they can launch a DoS attack by overloading the network with traffic, causing delays in transaction processing and block validation. Furthermore, they can act as miners in PoW, although this is very costly as it requires amassing enough computational power to control the network. In Delegated PoS, the attacker can use these identities to vote for themselves as a validator. In PBFT, the identities of nodes are typically known and trusted, making it difficult for Sybil identities to compromise the system. However, if the number of fake nodes exceeds one-third of the honest nodes, the system becomes vulnerable to this attack.

C. Denial of Service / Distributed Denial of Services

These attacks aim to make a network service unavailable to its users by overwhelming it with a flood of traffic, either from a single source (DoS) or from multiple sources (DDoS). Targeting the consensus mechanism by flooding the network with bogus requests or transactions, and specifically targeting nodes, miners, or validators, can cause serious damage to the network. It can slow down the network, therefore, delaying the processing of transactions. Additionally, the network's availability can be compromised by disrupting communication between nodes. It can even prevent nodes from participating by exhausting their resources.

V. Secure the Consensus in the Blockchained IoT

Securing consensus mechanisms tailored specifically for blockchained IoT environments is a critical aspect. In this section, we explore the adaptation of blockchain consensus mechanisms to IoT networks, as well as those designed specifically for IoT. We summarize the reviewed ones in Table I.

A consensus protocol called Proof of X-repute (PoX-R) [26] is designed to enhance the security and efficiency of public blockchain networks for IoT systems. It is a reputation-based protocol that combines Proof of X with a reputation layer to incentivize positive behaviour from participating nodes while discouraging malicious behaviour. The participating nodes are assigned reputation values based on their behaviour. They create a reward/punishment module to encourage/deter good/bad behaviour of participating nodes. For the consensus process, nodes with higher reputation values are given reduced difficulty in the mining process. This encourages nodes with positive contributions to publish more blocks. And the system will be more resistant to attacks and provide opportunities for nodes with lower computing power, to participate in the mining process. The proposed mechanism, however, makes use of a set of complex mathematical equations and a significant number of parameters to build a reward/punishment module. In addition, the network delay is neglected. Besides, there is a potential to use a significant amount of energy.

Proof-of-Authentication (PoAh) [27] consensus algorithm offers a secure and lightweight solution for resource-limited systems like IoT and edge computing. It is suitable for private networks. PoAh prioritizes cryptographic authentication over computational complexity, using the Elgamal cryptosystem for encryption and decryption. Devices on the network create blocks with digital signatures, sending them to the trusted node. Using public keys, the trusted node verifies and shares authenticated blocks across the chain. Successfully verified and authenticated blocks increase a node's trust value. In contrast, the trust value decreases if a fake block authentication is detected. By selecting trusted nodes, PoAh has significantly increased its security and eliminated the 51% attack. The algorithm's efficiency is demonstrated through theoretical analysis, simulations, and real-time test-bed deployment. This consensus, however, checks blocks using MAC and digital signatures, resulting in verified blocks holding erroneous data.

Yazdinejad et al. introduced SLPOW [28], Secure and Low Latency PoW, a consensus that draws inspiration from PoW. This paper tackles the limitations of centralized IoT systems, their limited scalability and security. They used a hash-based architecture to increase the security of IoT. This method defends the network against DoS attacks and a few other attacks. Besides bringing the computation of PoW onto a Field-programmable gate array (FPGA) to achieve high processing speeds of computation. Making it more secure and difficult for attackers to compromise the system, low latent, and energy-efficient. However, The paper neglects potential additional hardware costs for implementing the protocol.

Qian Qu et al. [29] proposed a novel Miner Twins (MinT) architecture to enable a fair PoW consensus for private blockchains in IoT environments. The MinT ensures security through continuous monitoring of miners via Digital Twins (DT) in the fog/cloud server, enabling the detection of misbehaving nodes. Anomalies are detected using Singular Spectrum Analysis (SSA) to identify individual miners violating fair mining policies, while the Proof-of-Behavior consensus algorithm detects colluding miners. However; the proposed MinT has only been evaluated in a proof-of-concept prototype implementation, and further testing is needed to validate their effectiveness in real-world scenarios. Additionally, the proposed SSA-based detection method may not be effective in all cases. Also, the proof of behavior consensus may not be able to detect all types of collusion.

Under the motivation of avoiding power-intensive calculations and securing Industrial IoT, Kara et al. [30] proposed a lightweight consensus protocol for IIoT that relies on chance to reach consensus instead of computing power. Based on certain randomized conditions, the chance value of a node will be updated at each consensus iteration and according to it, the node becomes a candidate. Then, if it has the chance again it will be selected as a miner to participate in consensus. The authors claimed this algorithm is lightweight, easy to use, flexible, and resistant to known attacks. PoCh tolerates more Byzantine faults than PoS by 20%, needing only 40% functional nodes compared to 50%. However, the limited number of candidates per block iteration weakens security and exposes the blockchain to attack. Additionally, the algorithm imposes various conditions that must be met and in case of failure to satisfy them results in a generated fail message.

TABLE I
SUMMARY OF REVIEWED PAPERS: ADDRESSED ISSUE, PROPOSED SOLUTION, MITIGATED ATTACKS, PROS AND CONS

Paper	Addressed Issue	Proposed Solution	M. Attack	Pros	Cons
Wang et al. [26] (2020)	Credibility of consensus protocol	Combines Proof of X with a reputation layer	Sybil 51%	Nodes with less computing power can participate	Complex equations, many parameters - Missed info. about delay & energy
Puthal etal. [27](2020)	Consensus is not secure & scalable for IoT	Cryptographic authentication for lightweight consensus	51%	Secure & lightweight solution - Low latency (3s)	Validated blocks with erroneous data.
Yazdinejad et al. [28](2020)	Limited security of IoT systems	The use of suitable hash method & FPGA	DoS - Increasing - Rectification - Dropping	Low latency Fast computation Energy efficiency	Hardware resource oversight Implementation cost
Qu et al. [29] (2021)	Unfair use of computing power by miners	MinT, a fair PoW using Digital Twins and SSA	Unfair mining	Fair mining Anomaly & collusion detection	Needs more evaluation and the proof of behavior may not detect all collusion
Kara et al. [30] (2022)	Power-intensive calculations and security	Selection of miners based on chance	Sybil - 51% DDoS Double spend	Lightweight & flexible Fault tolerance	Few candidates decrease security - Fail messages if conditions unmet
Sasikumar et al. [31](2022)	Centralized security of IIoT Slow block creation	Improved DPoS-based consensus for the IIoT network	N/F	Elevated data transactions rate - Energy consumption reduced	Disregarding many aspects & security criteria in evaluation
Kumar etal. [32](2022)	DLT-based IoT networks are vulnerable to DoS	Anomaly detection platform using ML & SW	DoS	High accuracy	Scalability
Liao et al. [33] (2023)	Reduced consensus efficiency and security in IoT with edge computing	Reputation and voting-based consensus mechanism	Collusion Byzantine	Good time consumption and transaction throughput	Increasing number of nodes - The system could centralize
Chen et al. [34] (2024)	Consensus security, efficiency and application	Node identity authentication and transformation protocol DAG-D, Double DAG, consensus	Sybil - DoS Double spend Eclipse	Efficiency High throughput Resistant to known attacks	Dependency on representative full nodes

Sasikumar et al. [31] have proposed an Improved DPoS consensus for improving the security of data analysis in an Industrial IoT (IIoT) network based on blockchain and AI for real-time data transmission. Their improved consensus relies on a mechanism of honour that is utilized to identify the malicious delegate nodes and subsequently replace them with honour delegate nodes. According to them, the proposed algorithm is more efficient than existing mechanisms when it comes to reliability, energy consumption, block generation speed, privacy, and security. However, the work was evaluated solely on energy consumption and transaction efficiency, neglecting many other aspects and security criteria.

To solve the vulnerability of DLT-based IoT networks to DoS attacks, Sathish et al. [32] have prototyped anomaly detection for IOTA-based IoT using machine learning and the Sliding Window (SW) technique. They have implemented this approach in the private IOTA network. The principle of this method is generating a security threat index for each DAG-based node by shifting the SW across the node's resource attributes, such as CPU and memory performance. The output of the SW is then fed into a multiclass classifier to obtain the threat index. The proposed approach achieved an accuracy rate of 98% and an F-measure of 88%. However, their study did not address the crucial aspect of scalability.

Liao and Cheng have developed RVC, a Reputation and Voting-based Consensus mechanism [33], to address the issues of security and efficiency in the consensus arising in edge computing-enabled IoT due to the increasing number of nodes. The approach begins with a simple mathematical calculation to determine the reputation value of participating nodes (servers and users). Thresholds are then calculated to filter out malicious nodes, followed by the selection of a block proposer. This proposer packages a new block and broadcasts it to the network. If 50% of the nodes vote for this block, it is added to the chain. This consensus demonstrates good results in terms of time consumption and transaction throughput. Additionally, it can resist malicious collusion between nodes and Byzantine attacks. However, an increasing number of nodes may create data storage issues for this consensus. Another concern is the system could centralize if the same proposer is selected each time based on its high reputation.

Consensus' security vulnerabilities, efficiency, and application have gained the interest of Chen Y. et al. [34], leading to the proposal of an efficient and secure consensus algorithm for Heterogeneous IIoT. Initially, they designed an IIoT system architecture based on Blockchain. To ensure consensus security, they proposed a node identity authentication and transformation protocol then classified nodes as full (representative and nonrepresentative), light and malicious according to it. In the next stage malicious nodes are prevented from accessing the network during the construction of the communication DAG. Based on this latter and transaction set DAG, a double-DAG consensus algorithm was proposed. Leveraging historical information, nodes vote on each transaction. Representative full nodes collect votes and assign different weights to them. Transactions are considered verified when enough votes surpass the threshold, while the entire transaction set is ready for the blockchain when over 2/3 of its transactions are verified. Despite its resistance to known attacks, double spend, DoS, Sybil and eclipse attacks, the system's reliance on representative full nodes for transaction propagation and verification poses a vulnerability if compromised.

VI. Conclusion and Future Works

This paper investigates the security of the consensus mechanisms for blockchain-based IoT systems. We have reviewed some consensus mechanisms and explored securing the consensus mechanisms for the Internet of Blockchained Things. Future research will aim to propose a solution for a secure and efficient consensus mechanism that respects as much as possible the constraints and specificities of the IoT.

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